AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) – 3

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Enrollment Number: A41105223039

Student Name: Anubhav Anand

Program: B.Tech CSE

Faculty Guide's Name: Dr. Girish Paliwal, Professor

Project Title: WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Introduction

Access to safe and adequate drinking water is recognized as a fundamental determinant of public health, yet a substantial proportion of India's rural population continues to rely on groundwater sources that are vulnerable to both geogenic and anthropogenic contamination. Uttar Pradesh and Bihar, two of India's most populous states, are particularly affected: extensive stretches of the Indo-Gangetic plain are known to harbour naturally occurring arsenic and fluoride contamination, compounded in places by nitrate and microbiological contamination arising from inadequate sanitation infrastructure. The Government of India's Jal Jeevan Mission (JJM), launched with the objective of providing functional household tap connections to every rural household, has simultaneously generated one of the largest water quality testing datasets in the country, capturing habitation-level results across a wide range of chemical and biological parameters.

Despite the scale and richness of this data, a significant gap exists between its collection and its practical use. Water quality test results recorded under JJM are primarily consolidated within the Water Quality Management Information System (WQMIS) and associated internal reporting layers, which are designed for national and state-level monitoring rather than for direct, actionable use by district and block-level field officials. Local administrators tasked with prioritising remediation activities — such as re-testing, source replacement, or treatment plant deployment — often lack a straightforward mechanism to convert raw contamination readings into a ranked, actionable list of intervention priorities. This creates a situation in which the data required to make evidence-based decisions exists, but is not structured in a form that directly supports those decisions.

This project is motivated by the observation that publicly accessible JJM water quality data can be independently processed to help close this gap, without requiring modification to, or privileged access within, existing government systems. Because the data spans multiple testing cycles between 2022 and 2025, it is possible not only to assess the severity of a single contamination reading but also to distinguish between persistent, recurring contamination and isolated anomalies — a distinction that carries direct implications for how urgently an intervention is required.

The primary objective of this project is to design and implement WaterTriage, an independently reproducible, open-source system that ingests publicly available JJM water quality data for Uttar Pradesh and Bihar, evaluates each recorded water source against Bureau of Indian Standards (BIS) 10500 permissible limits, and produces a ranked, filterable list of intervention priorities for use by local stakeholders. The system is scoped specifically to these two states both for practical reasons of data volume and computational feasibility within the project timeline, and because both states exhibit genuinely distinct contamination profiles — Bihar's arsenic-affected districts along the Gangetic floodplain contrasted against Uttar Pradesh's more widespread fluoride and nitrate concerns — offering a meaningful basis for comparative analysis.

The remainder of this report is organised as follows: subsequent sections review related work in water quality data systems and contamination risk scoring, describe the proposed methodology and system architecture, and present the results obtained from processing available data, followed by discussion, conclusions, and directions for future work.